

LabelCheck

How to Read Skincare Product Labels for Acne-Prone Skin

Sam | COMS W4170 | Spring 2026

Domain: Skincare

Topic: How to Read Skincare Product Labels for Acne-Prone Skin

This topic meets the requirements because it's a focused, practical skill that can be taught in 10 minutes through visual comparison, ingredient recognition, and short practice activities. It's specific enough to be learnable quickly, but difficult enough that many beginners do not already know how to do it confidently.

Expert: Sam

-**Sofia:** Uses skincare casually but does not know how to compare ingredient labels or tell which ingredients are most helpful for acne-prone skin.

-**Jolie:** Has seen common skincare ingredient names online but is not confident about what they do or how to choose between products.

Media: What media would you use?

1. Annotated product label images highlighting key ingredients for acne-prone skin. 2. Side-by-side product comparisons showing ingredient lists, active ingredients, and marketing claims. 3. Short interactive cards or visual callouts explaining what common ingredients do and when they are useful.

Positive TA Feedback: It seems like a topic a lot of people can relate to and it's very useful

Negative TA Feedback: You have to be able to narrow down the topic, enough for it to be easy to learn in a short amount of time

User:

- Harun Afza is a 20-year-old junior at Columbia College. They are interested in learning how to read skincare product labels because they want to choose products more confidently instead of relying on trends, packaging, or social media recommendations. For example, they often see ingredients like salicylic acid, niacinamide, or benzoyl peroxide on products, but do not know which ones are most relevant for acne-prone skin or how to compare two products while shopping.



Quiz:

1. Identify which ingredient in a label is most helpful for acne-prone skin. 2. Compare two skincare products and choose which one is more suitable for a beginner with acne-prone skin. 3. Distinguish between a meaningful ingredient claim and a vague marketing phrase on packaging

Who will you run your prototype on?

I'll run the prototype with another student in class who has used skincare products before but doesn't feel confident reading labels or comparing ingredients.

Skincare - Low-fidelity prototype

Idea: Basics of how to correctly choose skincare products for your skin

- **Person**

- Harun, a student who wants to improve their skincare routine due to acne

- **Goal**

- Teach them how to choose better skincare products

- **Subgoals**

- Recognize very common ingredients in skincare products
- How to read ingredient list
- Pass a final quiz

LabelCheck

Learn to read skincare product labels for acne-prone skin

Start Learning

Test Your Knowledge

Subgoal: User chooses to start learning or jump straight to the quiz. Default: Start Learning.

What You Are Looking For

When you pick up a skincare product, the front of the bottle shows you marketing. The back shows you facts. The ingredient list tells you exactly what is in the formula, listed from highest to lowest concentration.

In this lesson you will learn to identify 3 key ingredients that actually help acne-prone skin:

1. Salicylic Acid

Exfoliates inside the pore, unclogs blackheads and whiteheads

2. Niacinamide

Reduces redness, controls oil production, strengthens the skin barrier

3. Benzoyl Peroxide

Kills acne-causing bacteria directly on the skin surface

Learn: How to Read an Ingredient List

1

Order Matters

Ingredients are listed from highest to lowest concentration. If water is first, the product is mostly water. If an active ingredient appears near the end, there is very little of it.

2

Look for Actives in the Top Half

For a product to work, the active ingredient (salicylic acid, niacinamide, benzoyl peroxide) should appear in the first half of the list. If it is in the last few lines, the concentration is probably too low to do much.

3

Marketing Claims vs. Real Ingredients

Words like 'clarifying,' 'purifying,' or 'blemish-control' on the front label mean nothing on their own. Always flip the bottle and check the ingredient list to confirm.

Subgoal: User learns how to scan an ingredient list before reading about specific ingredients.

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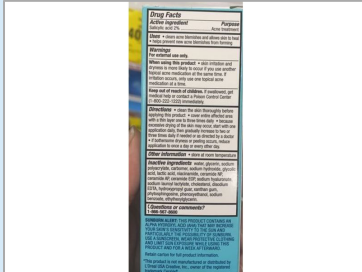
Learn: Salicylic Acid

GOOD: Present in top half of list

What it does: Dissolves the bonds between dead skin cells inside the pore, allowing them to exit naturally. It prevents blackheads and whiteheads from forming. Good for acne.

Best for: Oily or combination skin with clogged pores and occasional breakouts.

What to look for on a label: "Salicylic Acid 0.5%" to "2%"



WATCH OUT: Listed last or absent entirely

If a product says 'salicylic acid formula' on the front but salicylic acid appears in the last three lines of the ingredient list, the concentration is likely too low to make a real difference.

This is a common tactic on products that want to sound active without including enough of the ingredient to work.



What to look for: Find the ingredient and count how far down it sits. If it is past the halfway point, consider a different product.

Learn: Niacinamide

GOOD: 5% and above for real results

What it does: Reduces inflammation and redness, controls how much oil the skin produces, and helps repair the skin barrier. It is gentle enough for daily use.

Best for: Sensitive or inflamed acne, redness after breakouts, and oily skin.

What to look for: "Niacinamide" or "Vitamin B3" in the first half of the list.



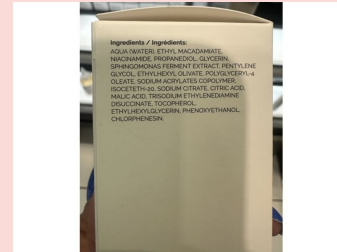
What to look for: The higher up niacinamide appears, the more effective the product is likely to be.

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WATCH OUT: Too low a concentration

Niacinamide needs to be at around 2% to 5% concentration to do anything meaningful. Many products add a tiny amount just to list it on the label.

If the label says "contains niacinamide" but does not list a percentage, and niacinamide appears near the end of the ingredient list, the amount is likely too low to help.



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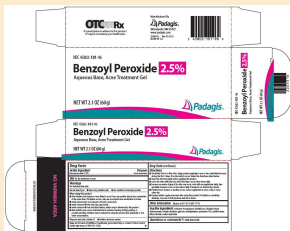
Learn: Benzoyl Peroxide

GOOD: Effective for active breakouts

What it does: Kills the bacteria (*Cutibacterium acnes*) that causes inflamed pimples. Works quickly on red, raised breakouts.

Best for: Active inflammatory acne, pimples that are red and swollen.

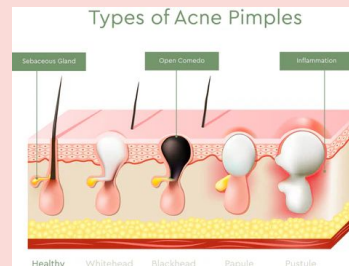
What to look for: "Benzoyl Peroxide 2.5%" or "5%" listed as the active ingredient, usually in its own section at the top of the label.



WATCH OUT: Wrong use case

Benzoyl peroxide doesn't really help with blackheads or clogged pores. That is salicylic acid's job. Using benzoyl peroxide on non-inflamed acne just dries out the skin without solving the problem.

It also bleaches fabric, so if you see it as an ingredient, keep it away from towels and pillowcases.



What to look for: Check whether your acne is inflamed (red, raised) or clogged (blackheads). Match the ingredient to the right type.

Quick Review

Ingredients are listed from highest to lowest concentration on every label

- 1. Salicylic Acid** Unclogs pores, prevents blackheads, look for it in the top half of the ingredient list
- 2. Niacinamide** Calms redness and controls oil, needs to be in the first half of the list at 2% or higher
- 3. Benzoyl Peroxide** Kills acne bacteria, best for inflamed pimples, not blackheads
- 4. Key Rule** Front-of-bottle claims mean nothing, always check the ingredient list on the back

Ready to test yourself? Let's go :))

Quiz: Question 1 of 3 — Check All That Apply

Look at this ingredient list. Which ingredients would actually help someone with acne-prone skin?

Ingredients

Active Ingredients:

Benzoyl Peroxide 2.5%

Inactive Ingredients:

Water, Ethoxydiglycol, Glycerin, Polyacrylamide, Chamomilla Recutita (Matricaria) Flower Extract, Aloe Barbadensis Leaf Juice, Allantoin, Niacinamide, Panthenol, Punica Granatum Fruit Extract, Yucca Glauca Root Extract, Ceramide AP, Ceramide EOP, Ceramide NP, Fragrance/Parfum, Biosaccharide Gum-1, Disodium Cocamido MIPA-Sulfosuccinate, Carbomer, Cholesterol, Leuconostoc/Radish Root Ferment Filtrate, Corn Starch Modified, Phytosphingosine, Sodium Lauroyl Lactylate, Xanthan Gum, Laureth-7, Aminomethyl Propanol, Disodium Oleamido MEA-Sulfosuccinate, C13-14 Isoparaffin, PEG-7 Glyceryl Cocoate, BHT, Disodium EDTA, Ethylhexylglycerin, Benzyl Alcohol, Phenoxyethanol, Citronellol, Geraniol, Limonene, Linalool.

☐

Salicylic Acid

☐

Fragrance

☐

Niacinamide

☐

Titanium Dioxide

☐

Benzoyl Peroxide

Submit

Subgoal: User scans a real ingredient list and identifies which ingredients actually help acne-prone skin.

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Correct: Salicylic Acid, Niacinamide, Benzoyl Peroxide

Here is how to read this list:

Benzoyl Peroxide: Listed as the active ingredient at 2.5%, so this is the main acne-fighting ingredient

Niacinamide: Present in the ingredient list and can be helpful for acne-prone skin

Salicylic Acid: Not present in the pictured ingredient list

Fragrance and Titanium Dioxide: These do not treat acne; fragrance may also irritate skin

Quiz: Question 2 of 3

You have two products. Which one is better for a beginner with acne-prone skin?

Product A

"Advanced Acne Solution"

Ingredients: Water, Alcohol Denat., Fragrance, Polysorbate 20, Aloe Vera Extract, Niacinamide, Salicylic Acid, EDTA, Methylparaben

Product B

"Daily Clear Gel"

Ingredients: Water, Glycerin, Niacinamide, Salicylic Acid, Sodium Hyaluronate, Allantoin, Carbomer, Sodium Hydroxide, Phenoxyethanol

A) Product A is better

B) Product B is better

C) Both are equally good

Subgoal: User compares two ingredient lists and picks the product with more effective active ingredients.

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Correct: B) Product B is better

The key clue: position of active ingredients and absence of irritants.

Product B: Niacinamide is listed third and Salicylic Acid is listed fourth. Both are near the top of a 9-ingredient list. No fragrance or high-concentration alcohol is present.

Product A: Alcohol Denat. is listed second (very high concentration), which can strip and irritate acne-prone skin. Fragrance is listed third. Niacinamide and Salicylic Acid both appear in the bottom half of the list.

Why not C? The same ingredients can be listed in very different positions, and irritating ingredients like alcohol and fragrance make Product A a worse choice regardless of what is on the front label.

Lesson: Two products can name the same ingredients but deliver very different results based on concentration order

Quiz: Question 3 of 3

True or False? Mark each statement.

1. *"Salicylic acid in the last three lines of an ingredient list is just as effective as salicylic acid listed third."*

T

F

2. *"The words 'purifying' or 'clarifying' on the front of a bottle confirm the product contains acne-fighting ingredients."*

T

F

3. *"Benzoyl peroxide is the right ingredient to use if you mainly have blackheads and clogged pores."*

T

F

4. *"Niacinamide listed near the top of an ingredient list is a sign the product contains a useful concentration."*

T

F

Submit

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Quiz: Answer — Question 3

1. FALSE "Salicylic acid listed last is just as effective."

Ingredient order reflects concentration. Listed last means the amount is likely too small to exfoliate or unclog pores effectively.

2. FALSE "Purifying or clarifying on the label confirms active ingredients."

These are marketing words with no regulatory definition. Always check the ingredient list to confirm what is actually in the formula.

3. FALSE "Benzoyl peroxide is right for blackheads and clogged pores."

Benzoyl peroxide targets bacteria that cause inflamed pimples. Salicylic acid is the correct ingredient for clogged pores and blackheads.

4. TRUE "Niacinamide near the top means a useful concentration."

Correct. An ingredient listed near the top is present in a higher amount, making it more likely to deliver a real benefit at 2% or above.

User Testing Insights

Tested on: Harun Afza (20, junior at Columbia College, uses skincare products but does not feel confident reading ingredient labels or comparing products)

[Positive] Harun immediately understood the ingredient order rule. They said it was a concept no one had ever explained to them before, and called it 'the most useful thing I've learned about skincare.' The side-by-side GOOD vs. WATCH OUT format made the pattern click quickly.

[Negative] On Quiz Q1, Harun hesitated for a long time over whether Benzoyl Peroxide counted as a correct answer. The question wording 'would help someone' was ambiguous. A clearer prompt like 'which are present in a useful amount?' would reduce confusion

[Positive] Harun found the product comparison quiz (Q2) the most engaging part. They said it felt like a real-world decision and asked if there could be more examples with products they had actually heard of.

[Negative] The distinction between Salicylic Acid and Benzoyl Peroxide was difficult to hold in memory. Harun mixed up which one targets blackheads vs. inflamed pimples during the quiz. A simple visual summary card showing 'SA = pores, BP = bacteria' would help before the quiz begins.

[Positive] Harun completed all 3 quiz questions and the full lesson. They got 2 out of 3 correct on the first attempt, which suggests the material is learnable but not really trivial. The topic difficulty was about right for a 10-minute lesson.

Was the user able to complete the prototype? Yes. Was the topic about the right difficulty? Yes, 2 out of 3 correct suggests the content is learnable but pretty much kinda challenging

Homework 9: Iterating on Low-Fidelity Prototypes:

Sofia Hirao - Basics of piping

What was at least one positive piece of feedback from your TA.

- She liked the interactive drag and drop feature that I have for the quiz and the feedback.

What is at least one negative piece of feedback from your TA.

- Make the instructions clearer right now it is just text on one side then pictures on the other integrate them together and for every visual step have written instructions that go along with it and don't cram everything on one slide use more slides.

Jolie Abadir - Squat Assessment

What was at least one positive piece of feedback from your TA.

- She liked the scope and overall format of the prototype and thought the topic was well suited for a 10 minute lesson.

What is at least one negative piece of feedback from your TA.

- Could use more interactive visual components like overlays and annotations to help the user understand what they are looking at rather than relying on text-based explanation.

Sam - Skincare basics

What was at least one positive piece of feedback from your TA.

-They liked the topic and the way the quiz was structured and also how i used real-life product examples to showcase the information

What is at least one negative piece of feedback from your TA.

-The pictures are too small and too much text

- I think I need to iterate my prototype more

Which two prototypes will you continue to iterate on for this week?

- 1) Sofia's Basics of piping
 - a) We will focus on iterating on the clarity and flow of instructions, integrating visual and textual steps together, and breaking down complex steps into separate slides so users can follow along more easily.
- 2) Jolie's Squat Assessment
 - a) We will focus on adding more interactive visual components like arrow overlays to help users identify compensation patterns and making the teaching content

more visually guided so the user can follow along without needing text-heavy explanation.